Assessment 1: ML Foundations

1. Data Management

Given that the dataset contains explicit patient labels, the objective is to predict discrete diagnostic categories (Depression, CI, and Both). Therefore, the problem is modelled as a supervised multi-class classification task.

Overview.

The clinical dataset comprises 670 patient records and 18 feature variables alongside a single target variable, with no missing values or duplicated instances (based on Patient Unique Identifier *id*).

The dataset features mixed numeric and categorical data with an unbalanced, three-class nominal target (Figure 1).

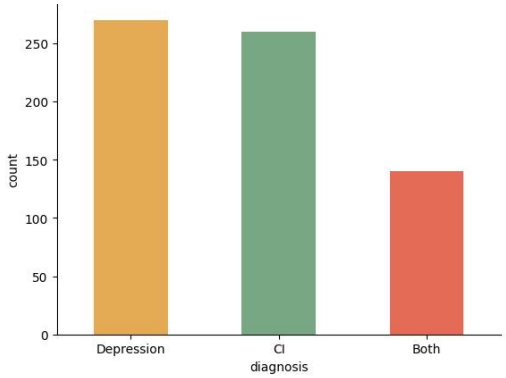

Figure 1. Distribution of the target variable *diagnosis*

Feature Removal.

Administrative data (*id*) and constant variables (*country*, all UK) were removed as they lack predictive validity for clinical diagnosis.

Distribution analysis and outlier detection.

Distribution analysis of numerical variables identified a severe outlier in the *depression phq9 score* (Figure 2). A record with a *depression phq9 score* of 99 was discarded, as it exceeded the clinical maximum of 27.

Also, the distributions of *depression phq9 score* and *fatigue severity scale score* have different peaks across different groups, implying potential important features to distinguish diagnosis.

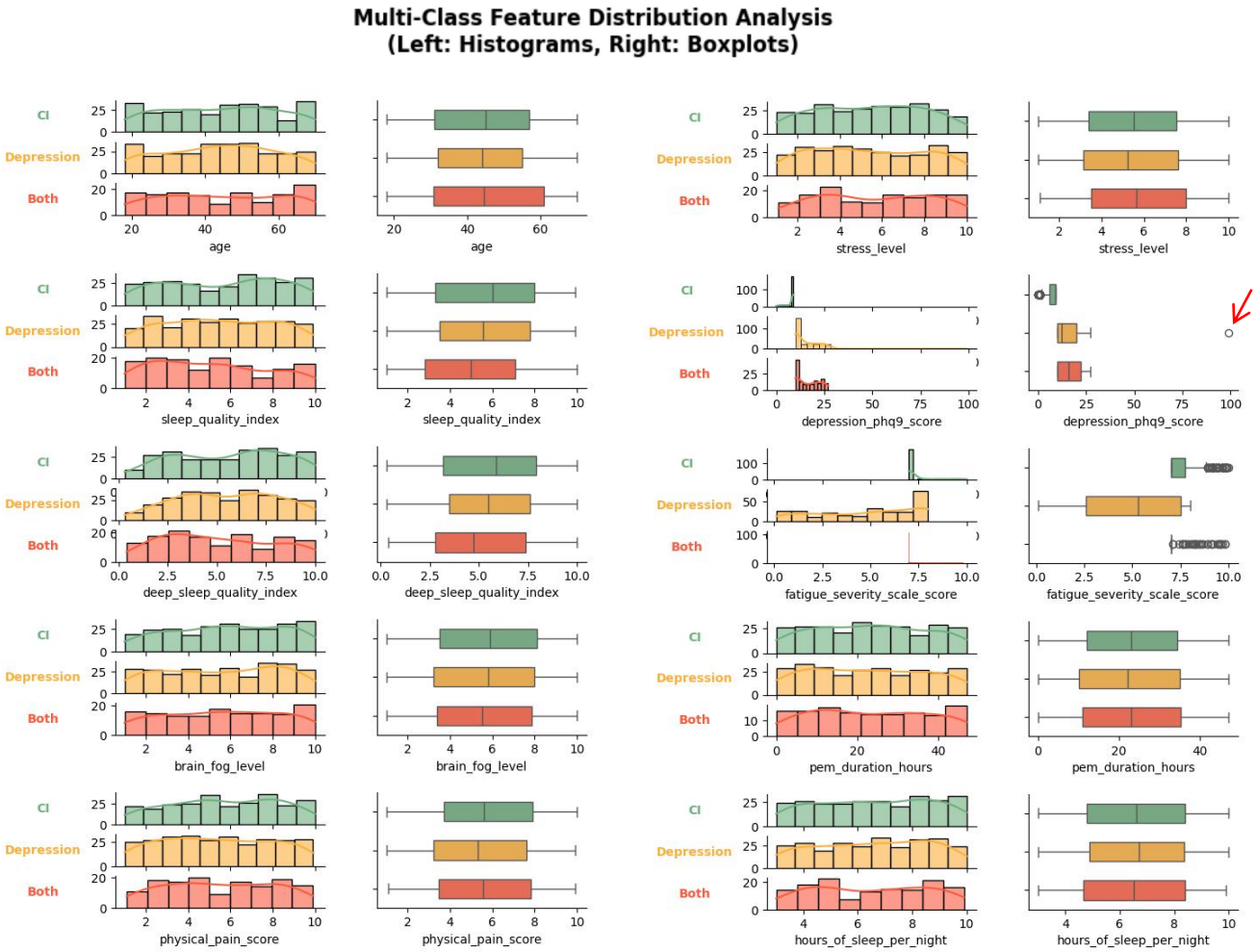

Figure 2. MultiClass feature distribution analysis (left: histogram, right: boxplot)

Correlations Analysis.

A perfect positive correlation (1.0) was identified between sleep quality index and deep sleep quality index (Figure 3). Consequently, the latter was eliminated to reduce redundancy.

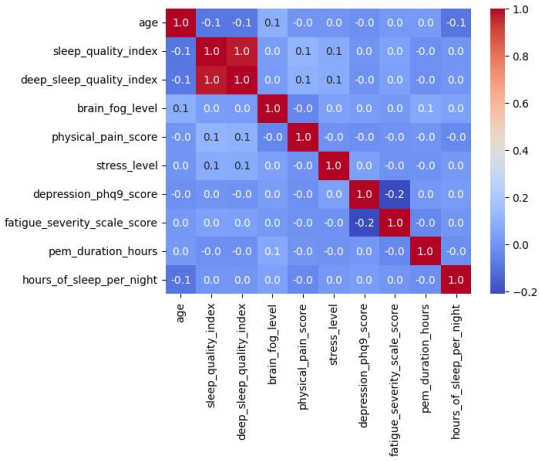

Figure 3. Correlation matrix of numerical variables

Encoding.

Ordinal features (*work status*, *social activity level*, *exercise frequency*) and the target variable (*diagnosis*) underwent label encoding, while nominal features (gender, *meditation or mindfulness*) were one-hot encoded.

Data split.

Finally, the dataset was partitioned into training and testing subsets using an 80/20 split. Stratification was used to preserve the original class distribution.

2. Model Training

Model Selection and Standardisation.

Three classifiers were selected: **Random Forest** (rule-based), **Logistic Regression** (probabilistic), and **Support Vector Machine** (distance-based). For the latter two, numerical features were standardized using Gaussian normalization within a pipeline to prevent data leakage.

Hyperparameter tuning.

A grid search with five-fold cross-validation was employed for three selected algorithms. Given the clinical diagnostic context of this study, macro-averaged recall was prioritized to minimize false negatives.

2.1 Random Forest

The hyperparameter exploration for the Random Forest model focused on three primary dimensions including the number of estimators, the maximum depth of the trees, and the minimum samples required at each leaf node. Table 1 presents a representative summary of the search results.

Table 1. Representative Hyperparameter Search Results for Random Forest

Group	Hyperparameters Explored			Performance Metrics		
	n_estimators (10, 50, 100)	max_depth (None, 3, 10)	min_samples_leaf (2, 10, 50)	Macro-Recall	Macro-Precision	Macro-F1
1 (best *)	10	None	2	100.00%	100.00%	100.00%
2	10	None	50	98.50%	99.24%	98.82%
3	10	10	10	100.00%	100.00%	100.00%
4	10	3	10	99.70%	99.84%	99.77%
5	50	3	10	100.00%	100.00%	100.00%
6	100	3	10	100.00%	100.00%	100.00%

* The "best" parameters were algorithmically selected, though multiple configurations achieved identical saturated performance across all metrics.

The model frequently achieved perfect macro-averaged scores (100%), suggesting highly discriminative clinical features. The optimization pipeline identified “max depth: None, min samples leaf: 2, n estimators: 10” as the "best" parameters. Although practical applications should favor higher tree counts or depth limits for better robustness, this study utilized the direct pipeline output to ensure consistency and reproducibility.

The min samples leaf parameter appeared to play a key role in regulating model performance. Increasing it to 50 (groups 1 & 2) decreased the macro-F1 score, showing that excessive regularization prevents capturing subtle subgroup signatures. Meanwhile, a larger max depth (groups 3 & 4) improved complex feature detection. Lastly, increasing n estimators (number of trees) enhanced model stability

but reached performance saturation at 50 trees, providing no further accuracy or recall benefits (groups 4-6).

2.2 Logistic Regression Model

The hyperparameter exploration for the Logistic Regression Model focused on two primary dimensions including the regularization parameter C and the penalty. The saga solver was selected because it natively supports multinomial classification while remaining the only solver compatible with L1 regularization. This allows for effective feature selection and prevents overfitting in a multi-class task with a relatively small dataset. Table 2 presents a representative summary of the search results.

Table 2. Representative Hyperparameter Search Results for Logistic Regression Model

Group	Hyperparameters Explored		Performance Metrics		
	C (0.001, 0.01, 0.1, 1)	penalty (l1, l2)	Macro-Recall	Macro-Precision	Macro-F1
1 (best)	1	l1	94.93%	97.52%	95.90%
2	1	l2	90.17%	95.47%	91.72%
3	0.1	l1	88.67%	94.91%	90.26%
4	0.1	l2	87.21%	94.32%	88.89%

The inverse regularization strength C played a pivotal role in modulating performance. Reducing C from 1 to 0.1 led to a decline in all three performance scores, indicating that excessive regularization hinders the capture of subtle subgroup signatures. Furthermore, the l1 penalty consistently outperformed l2 across all groups, benefiting from its inherent feature-selection capabilities. The model reached peak efficacy at C=1 with an l1 penalty (group 1), achieving a maximum macro-recall of 94.93%.

2.3 Supporting Vector Model

The hyperparameter exploration for the Supporting Vector Model focused on two primary parameters including the regularization parameter C and the kernel coefficient gamma. Table 3 presents a representative summary of the search results.

Table 3. Representative Hyperparameter Search Results for Supporting Vector Model

Group	Hyperparameters Explored		Performance Metrics		
	C (0.1, 1, 10)	gamma (0.01, 0.1, 1)	Macro-Recall	Macro-Precision	Macro-F1
1 (best)	10	0.01	88.37%	94.77%	90.02%
2	1	scale	85.69%	93.78%	87.35%
3	10	scale	86.08%	88.30%	86.77%
4	1	0.01	82.84%	92.80%	84.21%

Optimal performance was achieved in group 1 (C=10, gamma=0.01), yielding a peak macro-recall of 88.37%. Conversely, reducing C or transitioning gamma to scale resulted in a performance decrement (e.g., macro-recall dropping to 82.84% in Group 4), indicating that excessive regularization or suboptimal kernel scaling hinders the detection of subtle subgroup patterns.

3. Evaluate models

The Random Forest classifier with configuration (max depth: None, min samples leaf: 2, n estimators: 10) was identified as the top-performing model. To provide a final, unbiased evaluation of its clinical utility, the model was tested on a previously unseen hold-out test set (comprising 20% of the original data). Three distinct metrics were selected for the final report: macro recall, macro precision and macro F1 score. The final results on the test set are summarised in Table 4.

Table 4. Performances of the Final Optimal Model on the Test Set

	Recall	Precision	F1-score	Support
CI	100%	100%	100%	52
Depression	100%	100%	100%	54
Both	100%	100%	100%	28
Macro Recall	100%			
Macro Precision		100%		
Macro F1-score			100%	

The confusion matrix (Figure 4) visually corroborates these findings. The perfect diagonal alignment confirms that all test instances for CI, Depression, and Both group were classified with 100% accuracy. The lack of off-diagonal entries indicates no diagnostic overlap or misclassification, even for the smaller comorbid class.

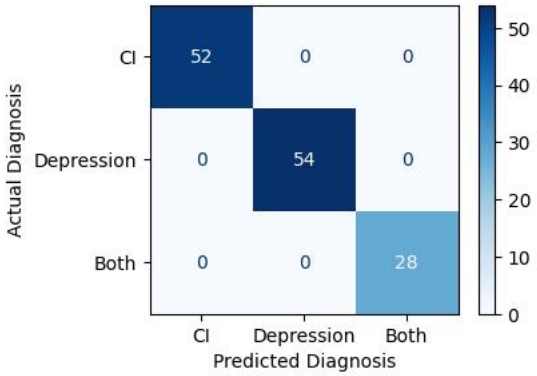

Figure 4. Confusion Matrix of the Final Optimal Model on the Test Set

While the models achieved 100% performance metrics, such absolute results are extraordinarily rare in clinical practice and often suggest an idealized scenario. This outcome is likely due to the high discriminative power of specific clinical scales (e.g., *depression phq9 score* and *fatigue severity scale score*) . However, real-world clinical data is invariably noisier and more heterogeneous. Future work must validate these findings using larger, independent external cohorts to ensure the model's practical utility.